

# Multimodal Fake News Detection Using Transformer-Based Text Models and Deep Visual Networks

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CSE 499A — Senior Design Capstone Project

**Abstract**—Online misinformation is increasingly multimodal: a misleading claim is usually paired with an image that lends it false credibility, so text-only classifiers discard part of the available evidence. This work implements and compares seven classifiers and one fusion model for binary fake news detection on a 17,045-sample Fakeddit-derived corpus of paired titles and image URLs. Four text models (TF-IDF+LogReg, mBERT, BERT-base, XLM-RoBERTa) and three vision backbones (ViT-B/16, EfficientNet-B0, ResNet50) were trained under an identical stratified split, after which a late-fusion model averaged the softmax outputs of XLM-RoBERTa and ResNet50. BERT-base achieved the highest accuracy (85.66%) and EfficientNet-B0 the best visual result (82.08%). Fusion reached 78.91% accuracy but the highest recall of any transformer pipeline (88.10%), trading precision for sensitivity to fake content.

**Index Terms**—fake news detection, multimodal learning, transformers, XLM-RoBERTa, ResNet50, late fusion.

## I. INTRODUCTION

Most deployed fake news detectors analyse only the textual claim, yet a large share of viral misinformation derives its persuasive power from an accompanying photograph that is manipulated, recycled, or shown out of context. This project therefore establishes strong single-modality baselines for both streams and quantifies what combining them contributes, through a Colab pipeline covering data preparation, training of seven models, late fusion, evaluation, and checkpoint export.

## II. DATASET AND PREPROCESSING

The corpus `final_train_17k.tsv` contains 17,045 samples derived from the Fakeddit benchmark, each providing a cleaned title, a hosted image URL, and a binary label (0 = authentic, 1 = fake). The classes are imbalanced at 10,404 authentic against 6,641 fake. A stratified split produced 13,636 training and 3,409 test samples, reused identically across every model so all numbers are comparable. Titles were tokenised with each model’s native tokeniser at a maximum length of 128 tokens. Images were fetched at run time, converted to RGB, resized to  $224 \times 224$ , and normalised with ImageNet statistics. Since some URLs are no longer reachable, a five-second timeout with a grey-image fallback was used; this preserves batch alignment but injects noise into the visual stream—a caveat for the image results.

## III. METHODOLOGY

**A. Text stream.** TF-IDF (5,000 features, unigrams and bigrams) with logistic regression was the classical baseline. Three encoders—mBERT, BERT-base-uncased, and XLM-RoBERTa—were fine-tuned with a two-class head using AdamW ( $\text{lr} = 2 \times 10^{-5}$ ), linear decay, gradient clipping at norm 1.0, and batch size 16; BERT variants for two epochs, XLM-RoBERTa for three.

**B. Visual stream.** ViT-B/16, EfficientNet-B0, and ResNet50, all ImageNet-pretrained, were adapted by replacing the final layer with a two-unit head and fine-tuned end-to-end for two epochs with Adam ( $\text{lr} = 1 \times 10^{-4}$ ), cross-entropy loss, and batch size 16.

**C. Late fusion.** Fusion operates at the decision level. For a sample with title  $t$  and image  $v$ , the softmax vectors of the fine-tuned XLM-RoBERTa and ResNet50 are averaged with equal weight:

$$\hat{y} = \arg \max_{c \in \{0,1\}} \frac{1}{2} [P_{\text{text}}(c | t) + P_{\text{img}}(c | v)]. \quad (1)$$

This keeps both encoders independently trainable and degrades gracefully when a modality is missing. All runs used an NVIDIA Tesla T4.

## IV. RESULTS AND DISCUSSION

Table 1 reports results on the 3,409-sample test set. The text stream dominates: every transformer outperforms every visual backbone, and TF-IDF is surpassed by roughly eight accuracy points, confirming that contextual embeddings capture rhetorical cues that bag-of-words features miss. The three transformers converge to within 0.8 points,

Table 1. Performance of all models on the test set (3,409 samples)

| Modality | Model           | Acc.          | F1            | Prec.         | Rec.          |
|----------|-----------------|---------------|---------------|---------------|---------------|
| Text     | TF-IDF + LogReg | 0.7715        | 0.6829        | 0.7431        | 0.6318        |
| Text     | mBERT           | 0.8542        | <b>0.8179</b> | 0.7966        | 0.8404        |
| Text     | BERT-base       | <b>0.8566</b> | 0.8177        | <b>0.8096</b> | 0.8261        |
| Text     | XLM-RoBERTa     | 0.8483        | 0.8047        | 0.8074        | 0.8020        |
| Image    | ViT-B/16        | 0.7580        | 0.6939        | 0.6840        | 0.7041        |
| Image    | EfficientNet-B0 | 0.8208        | 0.7967        | 0.7138        | <b>0.9014</b> |
| Image    | ResNet50        | 0.7870        | 0.7638        | 0.6724        | 0.8840        |
| Multi    | Late Fusion     | 0.7891        | 0.7650        | 0.6759        | 0.8810        |

so the multilingual capability of XLM-RoBERTa costs almost nothing while adding cross-lingual coverage—the reason it was chosen for deployment. The visual models show a different error profile, with EfficientNet-B0 and ResNet50 exceeding 0.88 recall at low precision, meaning they over-predict the fake class.

Fusion inherits that profile. Its 78.91% accuracy falls 5.9 points below XLM-RoBERTa because equal-weight averaging lets the weaker, high-recall image posterior override confident text predictions. Its 0.8810 recall is nevertheless the highest of any transformer configuration, suiting screening settings where a missed fake costs more than a false alarm; weighting the text posterior more heavily is the natural remedy.

## V. CONCLUSION AND FUTURE WORK

Seven classifiers and one late-fusion system were evaluated under a common protocol on 17,045 multimodal samples. BERT-base gave the best accuracy (85.66%), XLM-RoBERTa was adopted as the production text encoder, and EfficientNet-B0 was the best visual model (82.08%). Equal-weight fusion did not beat the best text model—an informative negative result, since naive averaging helps only when the streams are comparably strong. Future work will adopt a validation-tuned weighting, cache images locally, train the visual backbones longer, and add cross-attention fusion with LIME/SHAP explainability. Verified checkpoints are served through a Streamlit interface.

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